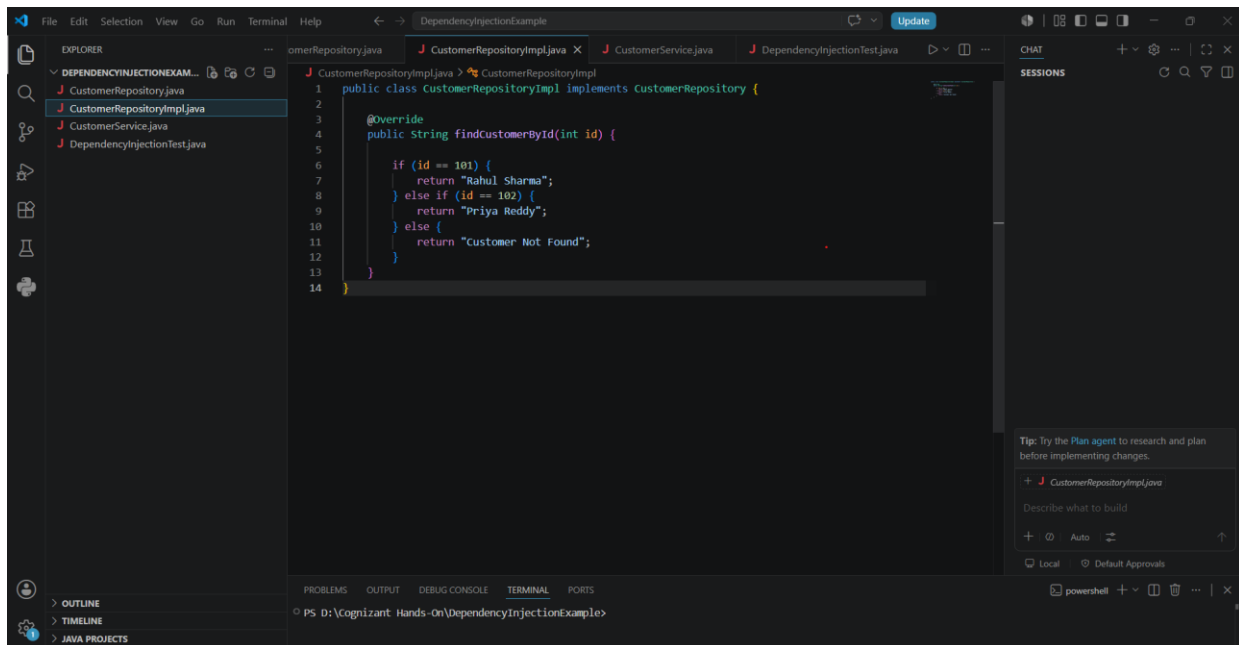


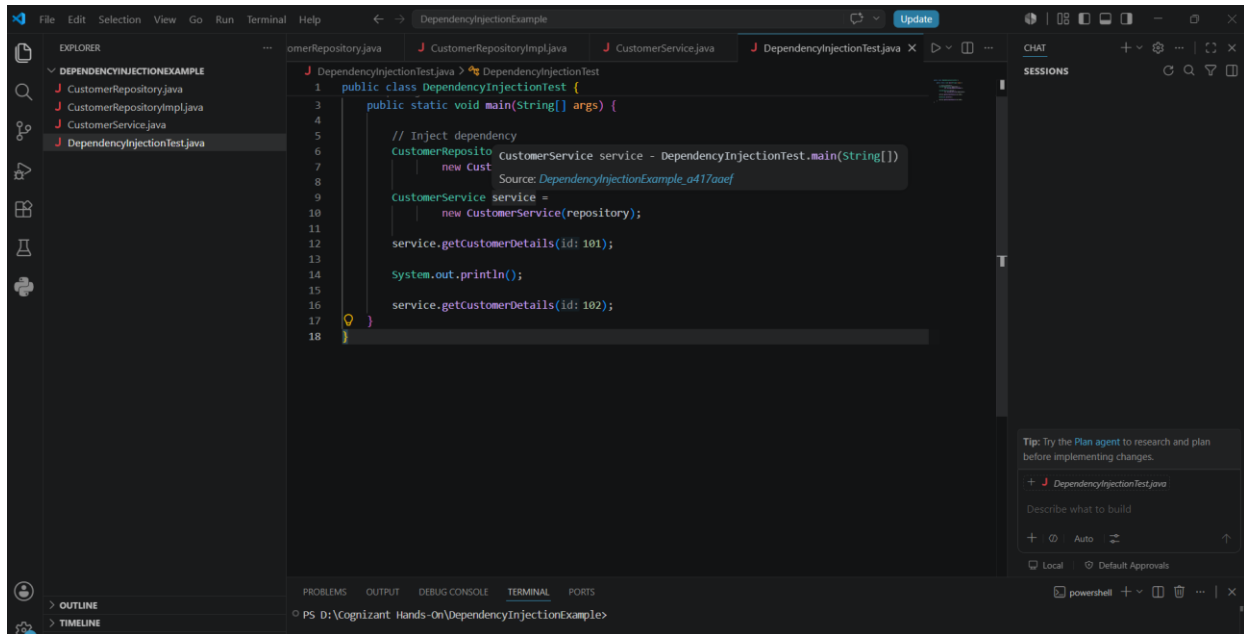
Hands-On

Exercise 11: Implementing Dependency Injection

In this module I have successfully developed, implemented and tested a project named `DependencyInjectionExample` using Dependency Injection in java. I defined an interface called `CustomerRepository` that has methods like `findCustomerById()` and wrote a concrete class called `CustomerRepositoryImpl` that implements the interface. I then created a `CustomerService` class that required a `CustomerRepository` interface instead of a particular implementation. To make the system loose coupled, I decided to use constructor-based Dependency Injection by passing the `CustomerRepository` as a dependency to the `CustomerService` class. Once implemented, I executed the app and verified the functionality by creating a `CustomerService` object, and calling the `CustomerRepositoryImpl` to fetch customer details. This exercise was very practical and I learned how Dependency Injection makes the application more flexible, maintainable, testable and follows the Dependency Inversion Principle by minimizing direct dependency between components.

Coding:





Output:

